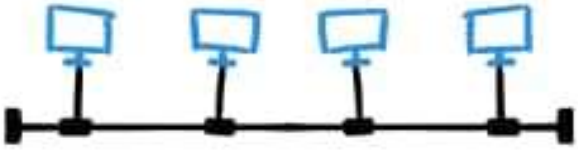
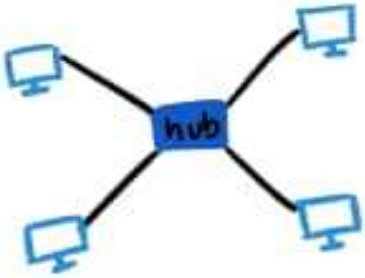


AsSiGnMenT oN UnIt IV

(network layer)







9th NOV 2024

Gayley Choden

NET 101

Computer Networks and Internet Protocols

Question 1

The aerial view of Gedu College of Business Studies is given below. The map includes the following landmark locations:



1. GCBS library-30hosts
2. GCBS (administrative building)-20hosts
3. Lecture theatre, GCBS-15hosts
4. GCBS GuestHouse-10hosts
5. GCBS prayerHall-10hosts

Assume that you are the ICT officer at GCBS tasked with providing internet access to all the areas above for the given number of hosts. The IP block provided to you is in the following format: 192.168.40.0/24.

Identify

a. Network and Broadcast ID
b. IP range of usable hosts
c. A neat network diagram for the solution

Solution

Subnet	1	2	4	8	16	32	64	128	256
Host	256	128	64	32	16	8	4	2	1
Subnet Mask	/24	/25	/26	/27	/28	/29	/30	/31	/32

Using this table I got from one of the references I found:

1. GCBS library- 30hosts = 32 host (taking the closest host)
2. GCBS (administrative building)- 20hosts = 32 host
3. Lecture theatre, GCBS- 15hosts = 16host
4. GCBS Guest House- 10hosts = 16host
5. GCBS Prayer Hall- 10hosts = 16host

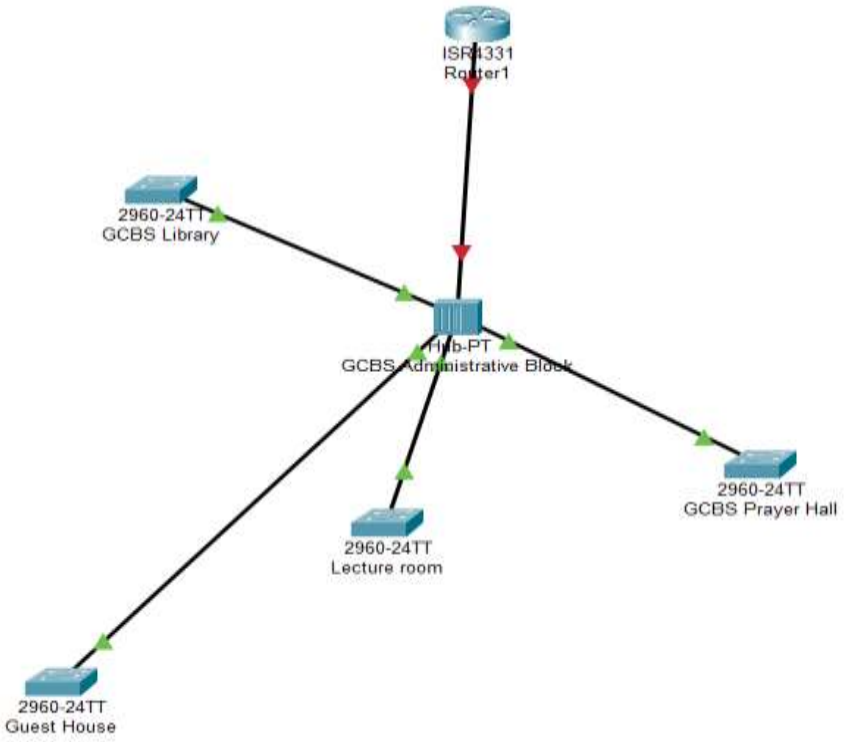
a.

Landmark	Network ID	Broadcast ID
GCBS library	192.168.40.0	192.168.40.31
GCBS (administrative building)-	192.168.40.32	192.168.40.63
Lecture theatre	192.168.40.64	192.168.40.79
GCBS Guest House	192.168.40.80	192.168.40.95
GCBS Prayer Hall	192.168.40.96	192.168.40.111

b.

Landmark	IP range of usable hosts
GCBS library	192.168.40.1 - 192.168.40.30
GCBS (administrative building)-	192.168.40.33 - 192.168.40.63
Lecture theatre	192.168.40.65 - 192.168.40.78
GCBS Guest House	192.168.40.81 - 192.168.40.94
GCBS Prayer Hall	192.168.40.197 - 192.168.40.110

c.

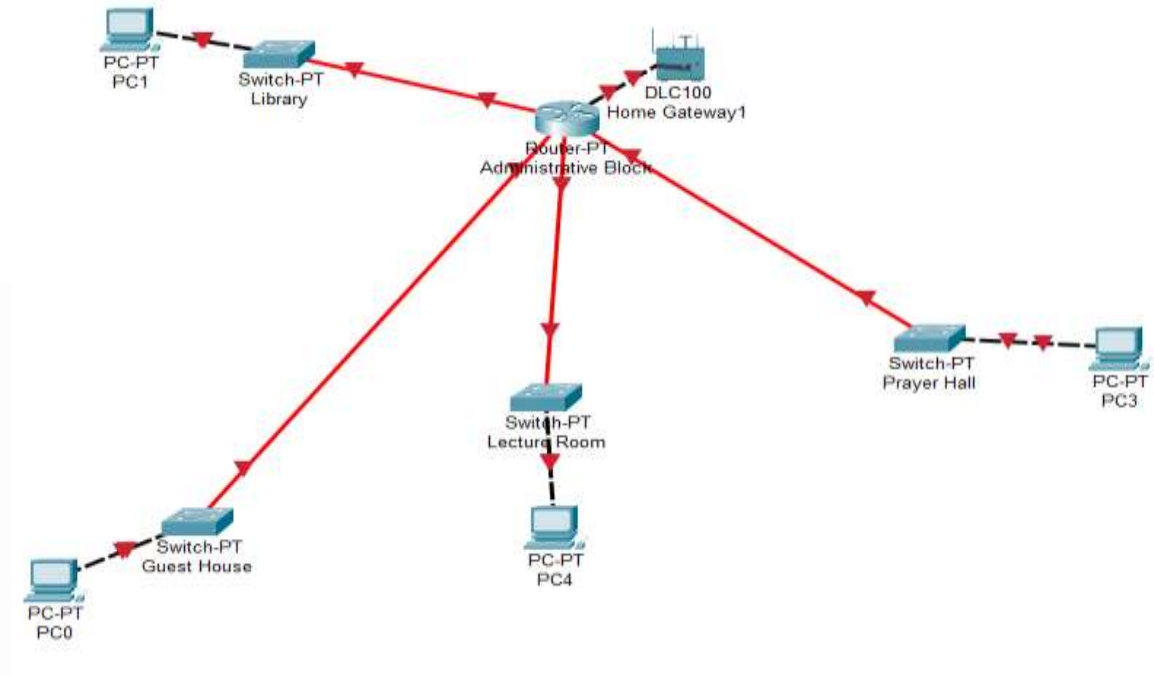


Question 2

While identifying the logical architecture of networks is important, it is also essential to understand the physical infrastructure associated with computer networks. For this reason, using the outline of the map given in Q1, use the following network devices and cables where necessary and redraw a rough sketch of the network layout. Justify the use of each of the following components:

1. Router
2. Switch
3. Gateway
4. Fiber Optic-cables
5. Twisted pair-cables

Solution



Router: Used router to connect the internal networks to external networks, acting as a gateway for data transfer between the college network and other networks.

Gateway: Provides a path for data to flow between different parts of the network and enabling external connectivity.

Switch: Used this switch to manage data and transfer data package from one device to the intended device (Library, Lecture Rooms, Guest House, Prayer Hall).

Fiber Optic Cables: Used switches to provide high-speed and high-capacity connections, which are especially important for the links between long distance buildings.

Twisted Pair Cables: Used to connect individual PCs to switches within each building. They are cost-effective for short distance.

Question3

Identify the most ideal network topology for your solution from question1 and justify your choice.

Solution

The most Ideal network topology would be Star Topology as the switches are connected to the router at the administrative block in the middle, creating a star-like structure. And Each building has its own switch, which connects the devices to the network. This setup improves reliability because even if one of the link fails, only that building is affected, while the rest of the network continues to operate normally. This centralized structure makes it easier to monitor, manage, and troubleshoot the network from a single location.

REFRENCES

https://youtu.be/YZRQpZPp4E4?si=KsGOHTR5QaPgICzh

https://youtu.be/ecCuyq-Wprc?si=aXLhfpBgihHllBO2

https://chatgpt.com/?model=auto